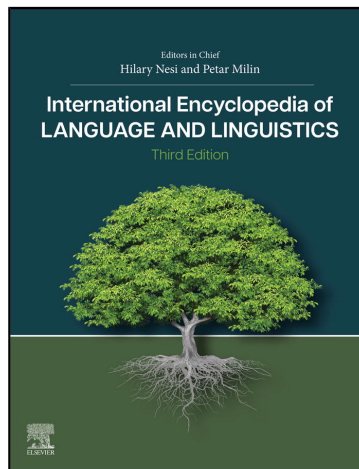


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McCrae, J. P. (2026). OntoLex-Lemon Model for Linked Data Lexicography. In: Nesi, H., Milin, P. (Eds.), *International Encyclopedia of Language and Linguistics*, 3e. Vol 10., pp. 747–753. UK: Elsevier.

<https://dx.doi.org/10.1016/B978-0-323-95504-1.01030-9>.

ISBN: 9780323955041

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OntoLex-Lemon Model for Linked Data Lexicography

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Key Points

- OntoLex-Lemon is a model for representing lexicographic data relative to ontologies and lexical concepts
- In addition, to the core module there are modules for syntax and semantics, decomposition, variation and translation, metadata, lexicography, morphology and frequency, attestation and corpus information
- The model has been used for a number of applications and we describe the applications to wordnets and to Wikidata

Abstract

OntoLex-Lemon is a W3C vocabulary for representing lexical information in RDF relative to ontologies. It consists of a number of modules allowing representation of different kinds of lexical information and has been adopted for a number of lexicographic resources, especially those based on open-source practices such as Open English WordNet and Wikidata.

List of Abbreviation

- FrAC Frequency, Attestation and Corpus Information
- LiME Linguistic Metadata Module
- OWL Web Ontology Language
- RDF Resource Description Framework
- SKOS Simple Knowledge Organization System
- W3C World Wide Web Consortium

Introduction

The OntoLex-Lemon Model (Cimiano et al., 2016; McCrae et al., 2012) is a W3C vocabulary for the representation of lexicographic data using the RDF (Wood et al., 2014) data model. The model aims to connect dictionary information with ontologies in the Web Ontology Language (Rudolph et al., 2012) and, thus, use ontologies in order to represent the meaning of words. Along with the core model a number of modules were released, namely modules for *syntax and semantics*, *decomposition*, *variation and translation* and

metadata. The community group has continued to develop modules with extra modules for *lexicography*, *morphology* and *frequency*, *attestation* and *corpus information* having been developed. The model has been adopted by a number of projects including BabelNet (Navigli & Ponzetto, 2010), Open English WordNet (McCrae et al., 2019, 2020) and Wikidata. In this article, the model, its modules and these applications are discussed.

Core Model

The core model of OntoLex-Lemon (Fig. 1) is based around the *lexical entry* as the key element in the model. A lexical entry collects all the forms and senses of a word that have a single morpho-syntactic pattern and set of senses and so can be used a single element for computational analysis. It is defined in the spec as follows:

A lexical entry represents a unit of analysis of the lexicon that consists of a set of forms that are grammatically related and a set of base meanings that are associated with all of these forms. Thus, a lexical entry is a word, multiword expression or affix with a single part-of-speech, morphological pattern, etymology and set of senses.

In order to enable computational linguistic applications of the model, lexical entries are not allowed to have multiple part-of-speech values, as they would have contradictory morphological information. Similarly, all senses should be representable by forms of the word, however we allow restrictions on the senses, for example allowing senses that can only be or cannot be in the plural. The lexical entry is composed of two key elements, the *form* and the *lexical sense*. The form represents one particular form that should have a single string representation (allowing for spelling or script variations) and can be uniquely described by a set of specific morphosyntactic properties, such as number, gender or tense. The lexical sense is notably different to senses in other lexical models, in that it describes the link to an ontological concept rather than being a meaning of its own. This means that the lexical sense does not generally have a definition as this would be superfluous to the definition given of the concept in the ontology. As such a simple entry in the Turtle serialization (Prud'hommeaux & Carothers, 2014) is as follows:

```
:cat a ontolex:LexicalEntry;
  ontolex:canonicalForm [
    ontolex:writtenRep "cat"@en
  ];
  ontolex:sense [
    ontolex:reference wikidata:Q146
  ].
```

Several properties are provided that further extend the possible modelling of lexical information. One form of the entry may be indicated as the *canonical form*, which indicates that it is the headword form used for the entry. The representations of forms can

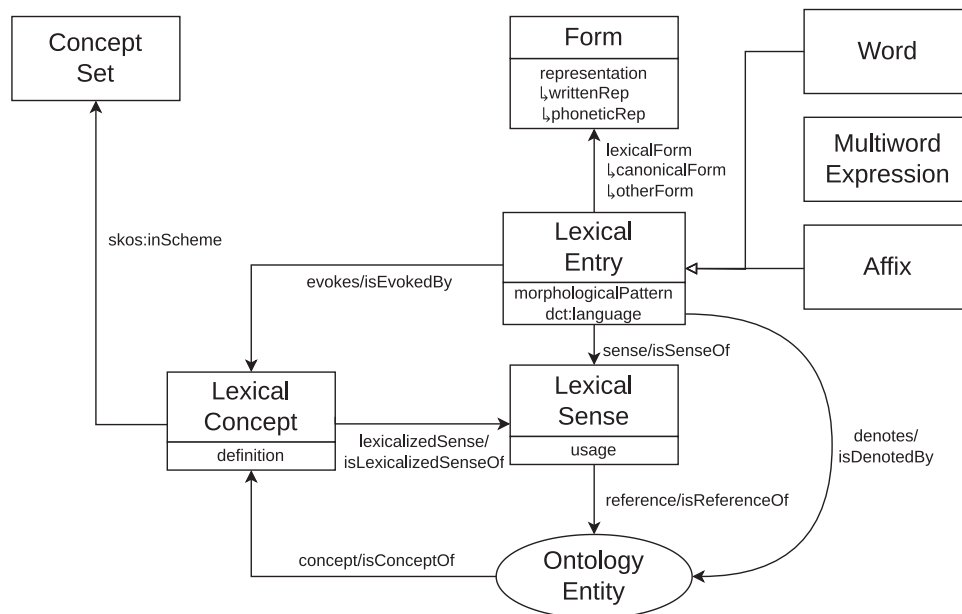


Fig. 1 The OntoLex-Core model.

be either *written* or *phonetic*, which is intended to give pronunciation information in a method such as the International Phonetic Alphabet (MacMahon, 2006). Further specializations of these properties and any other properties in the model are allowed. A lexical entry may be indicated to be either a *word*, *multiword expression* or *affix*. Further, the property *denotes* allows a direct link between the entry and the ontology entity to be specified, and there are axioms to supporting inference of this property with OWL reasoners. This means that the lexical sense may be omitted entirely, however if it is omitted it is not possible to give any description of the mapping between the entry its meanings in the ontology. A *usage* property is available on the lexical sense to describe this mapping. For example, we may indicate that “bulrushes” is a British English term as follows:

```
:bulrush a ontolex:Word;
  ontolex:sense [
    ontolex:usage [
      rdf:value "British English"@en
    ];
    ontolex:reference wikidata:Q147520
  ].
```

Finally, the *lexical concept* object was introduced to enable representation of lexicographic information without an ontology, which was not allowed in the previous *lemon* model (McCrae et al., 2012). This entry may have a definition unlike a lexical sense and it is assumed that this is used to define the meaning of the word in place of a formal ontology. As such, a lexical entry may *evoke* a lexical concept, while also *denoting* an ontology concept and the lexical concept may also be linked to the lexical sense, when these meanings are coincident. The choice of using a lexical concept as opposed to a lexical sense and ontology is generally defined by the semantic model: lexical concepts are typically defined by natural language definitions in the same language as the lexicon, where are ontology entities are language-independent and defined as RDF/OWL classes, properties or individuals.

Modules

Syntax and Semantics

The syntax and semantics module is used to describe how syntactic structures in natural language can be mapped to a sequence of triples based on the ontology. For example, the sentence “John knows Philipp” would need to be interpreted with individuals for “John” and “Philipp” and a property that corresponds to “know”. The main challenge is thus describing the syntactic frames and which properties are used as well as the ordering of the arguments in these properties.

The first step is the representation of the syntactic frame, which can be attached to the lexical entry and can use properties such as *subject* or *direct object* to represent the frame. For example, the following listing describes a transitive verb “know”:

```
:know a ontolex:Word;
  synsem:syntacticFrame [
    lexinfo:subject :know-subject;
    lexinfo:directObject :know-dirObj
  ];
  ontolex:sense :know-sense.
```

This is achieved by means of an “ontology map” object, which is attached to or may even be the same entity as the lexical sense. This maps the arguments defined in the syntactic frame to the *subject* (first position), *object* (third position) of the triple or to membership of a class. For example, the sense and ontology map for the above is as follows:

```
:know-sense a ontolex:LexicalSense, synsem:OntoMap;
  ontolex:reference foaf:knows;
  synsem:subjOfProp :know-subject;
  synsem:objOfProp :know-dirObj.
```

More complicated mappings can be described by composing multiple mappings together and restrictions on the mapping may also be defined in the ontology map to further enable the mapping of complicated elements.

Decomposition

The decomposition module describes how entries are composed, including the words that make up multiword expressions, as well as describing component words and affixes used in compound words. The key property here is the *subterm* property that links an entry to another entry that it is composed of. In addition to this, the model allows components to be defined which represent this same compositional relationship, but can be further annotated with grammatical information about the particular form used in the compound. This can even be extended to represent parse information or other complex representation of multiword expressions if required. For example, the term “applied linguistics” may be modelled as follows, where “applied” is considered to be a past participle.

```

:applied_linguistics a ontolex:MultiwordExpression;
  decomp:constituent :applied_comp, :linguistics_comp.
:linguistics_comp a decomp:Component;
  decomp:constituent :linguistics.
:applied a decomp:Component;
  decomp:constituent :apply;
  lexinfo:tense lexinfo:past;
  lexinfo:verbFormMood lexinfo:participle.

```

Variation and Translation

The variation and translation module describes the relationships between entries, senses and other elements in the lexicon. In this module, translation is seen as special case of variation between senses, where the senses are from different language lexicons. The module distinguishes between *lexical relations*, which do not refer to the sense of the word, and *sense relations*, which refer to a specific meaning of a word. Note that relationships that do not depend on a specific word such as broader/narrower relationships are modelled in other vocabularies such as OWL or SKOS. Relations can be represented as simple links, i.e., a single triple, or as objects that can carry further annotations such as provenance. As such, translation between two entries can be represented in the following way

1. By referring to the same entity in the ontology or to the same lexical concept;
2. By referring to two entities that are marked as equivalent ontological entities or lexical concepts;
3. As a direct link between the senses of two entries;
4. With a translation object that links the senses of two entries;
5. Directly between the two entries by means of the *translatable as* property.

The first two options are in practice the easiest as this means the translation holds between all entries with the same meaning. The third and fourth option can be used if you want to limit the translation to a particular word, for example if you wanted to provide some provenance information about an actual translation. The fifth option ignores the semantics of the word, and only indicates that in some contexts a word may be translatable to another word.

Metadata

The Linguistic Metadata (LiME) module of the allows description of the information contained in the lexicon and its linking to the ontology. This includes basic properties counting the number of lexical entries, lexical concepts and references to the ontology, as well as statistics about the average number of references per lexical entry or per reference. The module introduces key elements including the *lexicon* itself, as well as *lexicalization sets* and *conceptualization sets*, that contain all the links between a particular lexicon and its ontology or concept set, respectively. These can be further described with properties, including the language and other properties from Dublin Core or other metadata vocabularies. Finally, the model allows for partitions of a lexicon to be created and described individually.

Lexicography

The lexicography module (along with the morphology and FrAC modules), was not released as part of the original specification of the OntoLex-Lemon model, but was developed later and released as its own specification. This module was introduced to provide extra modelling to better capture dictionaries developed by traditional lexicographic practices. In particular, the model introduces a new (lexicographical) *entry* object, which does not have the restrictions on use of the lexical entry. In particular an entry can have multiple part-of-speech values or even be defined as only containing certain lexical senses. The entry can also be divided into sub-components to support hierarchical entry definitions found in many traditional dictionaries. The lexicographical entry is linked to the lexical entries, which can give fine-grained morphosyntactic information in a way that is consistent and interoperable with the rest of the module. For example, we could have an entry for the “*inglés*” as an adjective and a noun as follows:

```

:ingles a lexicog:Entry;
  lexicog:entry :ingles-noun, :ingles-adj.
:ingles-noun a ontolex:LexicalEntry;
  lexinfo:partOfSpeech lexinfo:noun;
  ontolex:canonicalForm [
    ontolex:writtenRep "inglés"@es.
  ].
:ingles-adj a ontolex:LexicalEntry;
  lexinfo:partOfSpeech lexinfo:adjective;
  ontolex:canonicalForm [
    ontolex:writtenRep "inglés"@es.
  ].

```

In addition, to the lexicographical entry, this module also introduces two useful modelling elements for *form restrictions* on senses, which limit the forms that can be used with a particular sense, for example, by saying that some senses only apply to plural forms of the entry. A second useful addition is the inclusion of a formal *usage example* object to give examples of usage of an entry.

Morphology

The morphology module describes the inflectional, derivational and compounding morphology of entries in the lexicon, both by providing segmentation of forms of the lexical entries and also by describing the morphological rules that apply to forms. As such, the model can not only describe the morphology of existing forms but also generate new forms based on grammatical information such as the declension or conjugation of the entry.

The core of the morphology is the *morph*, which is a special kind of lexical entry and unlike normal entries forms in the lexicon can be composed of morphs. These morphs are also allowing to have a *grammatical meaning*, which describes the interpretation of the morph, as opposed to its actual sense in the ontology. This methodology allows the form to be decomposed into morphs and there are several subclasses of morphs that describe specific morphological types such as prefixing and suffixing. The morph may also be associated with a number of rules, that describe how the morph can be used. These can be simple replacement rules describing the morphological process in terms of replacing part of the string with another string, but more complex systems may also be described. In addition, to describing the composition of forms, there are also *word formation rules*, which are a special kind of lexical relation and enables the description of derivational rules between lexical entries. For example, the following models the derivation of the German word “Schönheit”:

```
:schoenheit-entry a ontolex:LexicalEntry;
  ontolex:canonicalForm [
    ontolex:writtenRep "Schönheit"@de ].

:schoen-entry a ontolex:LexicalEntry;
  ontolex:canonicalForm [
    ontolex:writtenRep "schön"@de ].

:schoen_heit a morph:WordFormationRelation;
  vartrans:source :schoen-entry;
  vartrans:target :schoenheit-entry;
  morph:wordFormationRule :_heit-rule.

:_heit-rule a morph:WordFormationRule ,
  morph:DerivationRule;
  morph:involves :_heit-morph;
  morph:generates :schoenheit-entry.

:_heit-morph a morph:Morph, morph:Suffix;
  ontolex:lexicalForm [
    ontolex:writtenRep "-heit"@de ].
```

Frequency, Attestation and Corpus Information

The Frequency, Attestation and Corpus Information (FrAC) module is concerned with connecting lexicons with corpus resources and thus to represent information about the entries. The model defines an *observable* element, which is one of the elements of the lexicon, such as a form, entry, sense or concept, and an *observation* of the element. The observation of the element could be a single *attestation* of the word in a corpus, which may be given with the text and a citation or by using the Web Annotation Data Model (Sanderson et al., 2017) to connect it to a resource on the web. Observations may also consist of statistics about the observable, and as such a *frequency* allows measurements of the number of occurrences of a form, entry or sense in corpus including information of how the count was obtained and on which data. A *collocation* is a further extension that provides frequency information about two or more observables, for example providing information about how often two lexical entries are found in the same sentence. Further, observation such as word embeddings or similarity scores may also be modelled as observations.

LexInfo

LexInfo (Cimiano et al., 2011) is not a formal part of the OntoLex-lemon model, but is an independent vocabulary that is developed by the OntoLex community group. LexInfo collects a large number of properties and values that can be used as standard data categories in order to model lexicons and can be used as a reference model for multiple lexicographic resources. LexInfo includes the following properties and values:

Morphosyntactic Properties

Properties and values for morphosyntactic properties such as part-of-speech, gender, number, tense.

Relations

Lexical and sense relations that can be used with the variation and translation module

Representations

Further properties for representing forms such as romanization and transliteration.

Definition and usages

More specific properties to describe the usage or definition of terms, including temporal or geographical usage and etymology

Syntactic arguments and frames

Syntactic arguments for use with the syntax and semantics modules and frame classes defined by the arguments used

Word types

More subclasses of lexical entry, including particular grammatical constructions such as a noun phrase or more specific affix classes such as prefix and suffix.

Miscellaneous

Other properties such as confidence and language-specific constraints on frames and arguments.

The LexInfo ontology is continually developed by contributions to the GitHub site and published as a single ontology.

Applications**WordNets**

The OntoLex model has been adopted by the Global WordNet Association as one of the serializations used for the Collaborative Interlingual Index (Bond et al., 2016) and thus adopted by major projects such as Open English WordNet (McCrae et al., 2019, 2020). The modelling in OntoLex is well-aligned with that of wordnets, with the synsets being treated as the ontology and the lexical entry and sense working as above. In addition, a small extra ontology has been created that includes the wordnet part-of-speech values and relationships such as *hypernymy*. In addition, specific properties such as the Interlingual Index Identifier are defined by this ontology. This model has thus been adopted by projects such as Open English WordNet and a linked data interface to this ontology at <https://en-word.net/>, which replaces the moribund Princeton WordNet.

Wikidata

Another large resource to adopt OntoLex is Wikidata (see Chapter X for more details on Wikidata and OntoLex), which is developing a database of lexemes based on the OntoLex-lemon model. The modelling adopted by Wikidata consists of lexemes (corresponding to lexical entries), forms and senses as in the main OntoLex model. The reference ontology used for this is naturally Wikidata's main item sets and they are linked to senses of the word by means of the "item for this sense" property corresponding to the reference in the OntoLex-lemon model. Lexemes in Wikidata represent all forms of the word with appropriate grammatical categories drawn again from Wikidata items. As such, a form may be annotated with Q146078 for the "accusative case." Senses may have definitions, although these are often missing as they are redundant to the definitions given on the items. Senses also have translations at the sense level and can have relationships between senses such as for antonyms.

Conclusion

The OntoLex-lemon model has been developed by a large community over a long period and has been adopted by a number of projects. It provides robust modelling for a range of lexicographic formalism and has expanded beyond its original goal of representing lexical information about ontologies to become a general model for representing lexicographic data using web standards. New modules are under continuous developments and further revisions to the model will enhance the usability of the model for a wide range of applications.

This work has been supported by Science Foundation Ireland under Grant Number SFI/12/RC/2289_P2 Insight_2, Insight SFI Centre for Data Analytics and the ADAPT Centre for Digital Content Technology (grant number 13/RC/2106 P2).

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